

Complements of Operational Research

7.2 Inventory management: planned shortages and quantity discounts

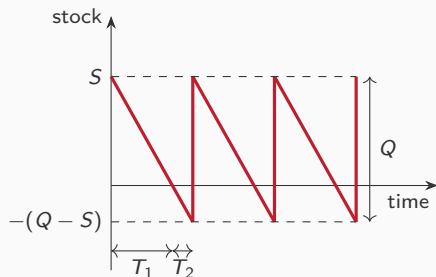
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Running out on purpose?

The factory sells a *premium* exterior paint online. Customers accept waiting a few days if there is no stock (**backorders**, filled as soon as the next order arrives).

Letting stock run out a little saves holding costs — but it has a cost: compensation discounts, express shipping, image.

- $D = 1200$ cans/year; $C_a = 50$ €; $C_e = 80$ €/order;
- holding 20%/year: $C_p = 10$ €/can/year;
- stockout: $C_r = 30$ €/backordered can/year;
- lead time: 1 week.



S : maximum stock; $Q - S$: maximum shortage;
 $T_1 = S/D$ with stock, $T_2 = (Q - S)/D$ out of stock.

The model with shortages

Formulas (the ones used in class)

Service parameter: $\rho = \frac{C_r}{C_p + C_r}$

$$Q^* = \sqrt{\frac{2 D C_e}{C_p}} \cdot \frac{1}{\sqrt{\rho}}, \quad S^* = \rho Q^*$$

$$K = D C_a + \frac{D}{Q} C_e + C_p \frac{S^2}{2Q} + C_r \frac{(Q - S)^2}{2Q}$$

Reorder point: $P_e = D \cdot \text{lead time} - (Q - S)$.

If $C_r \rightarrow \infty$ (running out is forbidden), $\rho \rightarrow 1$ and we are back to the EOQ of 7.1.

Reading ρ : optimal fraction of the cycle with positive stock, $T_1/T = S/Q$. In the class exercise statements it is called “service level”; in other texts, with random demand, service level means other things (e.g. the probability of no stockout in a cycle).

The cheaper it is to run out (C_r small) compared with storing (C_p), the longer we accept being out of stock.

Negative reorder point: we order when customers are already waiting — the order arrives in time to serve them without exceeding the maximum shortage.

Result: the *premium* paint

fraction with stock $\rho = 30/40$	75%	€/year	with shortages	no shortages
$Q^* = \sqrt{2 \cdot 1200 \cdot 80/10} / \sqrt{0.75}$	160 cans	ordering	600.00	692.82
maximum stock S^*	120	holding	450.00	692.82
maximum shortage $Q^* - S^*$	40	shortage	150.00	—
cycle T	1.6 months (6.9 wk)	sum	1200.00	1385.64
with stock / out of stock	5.2 / 1.7 weeks			
reorder point	$23.1 - 40 = -16.9$			

Policy: order 160 cans when there are 17 cans on backorder (about 7.5 orders per year).

Accepting planned shortages saves 185.64 €/year (13% of the management costs) — **if** customers accept waiting.

The purchase ($1200 \times 50 = 60\,000$ €) is the same in both cases; the total cost is 61 200 €.

Quantity discounts

The titanium dioxide supplier (7.1) offers:

Order (kg)	discount	price (€/kg)
less than 3000	—	4.00
3000 to 5999	3%	3.88
6000 or more	5%	3.80

Now the price depends on Q , and so does the holding cost ($C_p = 0.2 \times \text{price}$). The purchase **enters** the comparison.

Procedure

For each price break:

1. compute the EOQ with that bracket's price;
2. if it falls outside the bracket, use the nearest limit (in general, the bracket's minimum);
3. compute the total cost, *including the purchase*.

Choose the bracket with the lowest total cost.

Result of the discounts

Bracket	price	EOQ	Q used	purchase	ordering	holding	total (€/year)
< 3000	4.00	2121	2121	48 000	848.53	848.53	49 697.06
3000–5999	3.88	2154	3000	46 560	600.00	1164.00	48 324.00
≥ 6000	3.80	2176	6000	45 600	300.00	2280.00	48 180.00

Decision: order 6000 kg (two orders per year):
saves 1517 €/year compared with the EOQ without discount.

The difference from 3000 kg is only 144 €/year —
before deciding, check whether there is room for
6000 kg and whether the product does not
deteriorate.

In the discount brackets the EOQ (2154, 2176) is
below the bracket's minimum: the management
cost rises when ordering more, but the saving on
the purchase (1440 € and 2400 €) makes up for it.

It is a trade-off that only appears because the
purchase is no longer constant.

```
from math import sqrt

def rutura(D, Ce, Cp, Cr, Ca=0):
    rho = Cr / (Cp + Cr)
    Q = sqrt(2 * D * Ce / (Cp * rho)); S = rho * Q
    K = D*Ca + D*Ce/Q + Cp*S**2/(2*Q) + Cr*(Q - S)**2/(2*Q)
    return rho, Q, S, K

def descontos(D, Ce, I, escaloes):      # [(q_min, q_max, price)]
    melhor = None
    for lo, hi, Ca in escaloes:
        Q = min(max(sqrt(2 * D * Ce / (I * Ca)), lo), hi)
        K = D * Ca + D * Ce / Q + Q / 2 * I * Ca
        if melhor is None or K < melhor[1]: melhor = (Q, K)
    return melhor

print(rutura(1200, 80, 10, 30, 50))      # (0.75, 160, 120, 61200)
print(descontos(12000, 150, 0.2,
    [(0, 2999, 4.0), (3000, 5999, 3.88), (6000, 1e9, 3.8)])) # (6000, 48180)
```

With a spreadsheet: one row per price break and the MIN function.

Try it: from what discount in the last bracket would ordering 6000 kg no longer pay off?

Example 3 — AAA shop

The AAA shop sells a product XAA, which it buys from the company BBB at 3.20 € and sells at 5.25 €. It sells, on average, 21 units per week. The annual holding rate is 25% and the ordering cost is 20 €. BBB offers a 7% discount for orders of 400 to 899 units, 10% from 900 to 1999 and 15% for 2000 or more. What is the optimal order? (Year with 52 weeks.)

Example 4 — imported components

A company uses 15 000 components per year, at 100 m.u. each. Discounts: 5% from 1000 to 2999; 6% from 3000 to 5999; 9% from 6000 to 8999; 10% above 9000. Ordering: 150 m.u.; holding: 20%/year. Find the EOQ, the cycle, the annual cost, the number of orders per year and the optimal policy.

Exercise sheet 7 — Exercise 2 (homework)

Inventory management with shortages (homework)

Time unit: year. Demand: 600 units/year; holding rate 20%; unit purchase cost 300 m.u.; ordering cost 50 m.u.; service level (ρ) 95%.

- a) Economic order quantity? b) Active (maximum) stock? c) Stock cycle period?
- d) Sub-periods of positive and negative stock?
- e) Annual purchase, ordering, holding and shortage costs, and total cost.

Submission: by email to the lecturer, by the deadline given in class. All exercises: **Exercise sheet 7**.

KEY TAKEAWAYS — 7.2

1. Planned shortages: $\rho = C_r / (C_p + C_r)$ = fraction of the cycle with positive stock;
 $Q^* = \sqrt{2DC_e / C_p} / \sqrt{\rho}$; $S^* = \rho Q^*$; the reorder point can be negative.
2. Accepting shortages pays off when running out is cheap compared with storing — and customers wait.
3. Discounts: EOQ in each bracket, adjust to the bracket, compare the total cost *with* the purchase.
4. Always look at the margin: close decisions call for extra criteria (space, shelf life, risk).

Next: 7.3 When we produce it ourselves: the production and consumption model

References Full list: *Course bibliography*.

- Hillier, F. S., Lieberman, G. J., *Introduction to Operations Research*, 11th ed., McGraw-Hill, 2021 — Inventory Theory (planned shortages, quantity discounts).
- Taha, H. A., *Operations Research: An Introduction*, 10th ed., Pearson, 2017 — deterministic inventory models.
- Heizer, J., Render, B., Munson, C., *Operations Management*, Pearson — ch. 12.

Resources

- Exercise sheet 7. `code/python/07_inventory/07_2_shortages_discounts/rutura_descontos.py`.